DOBUTAMINE

High-alert drug

Inotrope, not vasopressor

CHEMICAL NAME (RS)-4-(2-{[4-(4-hydroxyphenyl)butan-2-yl]amino}ethyl)benzene-1,2-diol (hydrochloride)

CLASS Synthetic catecholamine inotrope (β1 agonist)

TRADE NAMES Dobutrex · Dobutamine HCl infusion · Inotrex

THERAPEUTIC USES / INDICATIONS

- Cardiogenic shock / low cardiac output (acute heart failure, after cardiac surgery).
- Septic shock with myocardial dysfunction (added to noradrenaline).
- Pharmacological stress echocardiography.

ADMINISTRATION

- **IV infusion** via syringe pump (central or large peripheral vein).
- Correct hypovolaemia first.
- Incompatible with alkaline solutions (bicarbonate).

DOSAGES

- **Adult:** 2.5–10 mcg/kg/min, titrated (up to 20 mcg/kg/min; rarely 40).
- **Children:** 2–20 mcg/kg/min.
- **Stress echo:** 5 mcg/kg/min, increasing every 3 min to 40 mcg/kg/min.

PHARMACOKINETICS

- **Onset:** 1–2 min; peak 10 min. **Duration:** <10 min after stopping.
- Metabolised by COMT in liver and tissues.
- Excreted in urine. **t½:** 2 min.

PHARMACODYNAMICS

- Predominantly **β1 agonist** → ↑ myocardial contractility and cardiac output.
- Mild β2 → vasodilation (↓ afterload); may ↓ BP.
- Less tachycardia than isoprenaline.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Tachycardia, palpitations.
- Headache, nausea.
- Hypotension (β2 vasodilation).

ADVERSE EFFECTS (SERIOUS)

- Arrhythmias (ventricular ectopics, AF).
- Myocardial ischaemia, angina.
- Hypokalaemia.
- Tachyphylaxis after 72 h.

CONTRAINDICATIONS & CAUTIONS

- Hypertrophic obstructive cardiomyopathy (outflow obstruction).
- Severe aortic stenosis.
- Uncorrected hypovolaemia.
- Sulfite hypersensitivity (some products).

KEY INTERACTIONS

- **β-blockers:** reduced effect.
- **MAOIs:** ↑ effect.
- **Volatile anaesthetics:** arrhythmias.

NURSING MANAGEMENT

- Continuous ECG, BP; monitor cardiac output indicators (urine output, lactate, peripheral perfusion, mental status).
- Monitor K^+.
- Watch for chest pain and arrhythmias.
- Titrate per haemodynamic targets; wean gradually.

PATIENT EDUCATION

- Explain purpose: helps the heart pump more strongly.

KEY POINTS

- β1 inotrope; ↑ cardiac output.
- May lower BP (β2) — not a vasopressor.
- 2.5–10 mcg/kg/min.
- Arrhythmias, ischaemia.

SOURCES CHECKED **[383]** Dobutamine 12.5 mg/ml concentrate for solution for infusion (UK SmPC) **[381]** Evans et al., Intensive Care Med 2021 **[384]** Dobutamine, IV (US label) **[385]** Dobutamine, IV (US label, Slate Run Pharmaceuticals)

Full references p. 432–457